

Søren Riis

AI Researcher and Mathematician

Reader in Computer Science, Queen Mary University of London

Kidlington, United Kingdom | s.riis@qmul.ac.uk | [Research website](#) | [LinkedIn](#) | [GitHub](#)

Profile

Mathematician and AI researcher with an Oxford DPhil and a research record spanning computational complexity, logic, information theory and reinforcement learning. Studies how AI systems learn mathematical structure and how to evaluate capability beyond competitive performance. Prize-winning contributor to Humanity's Last Exam, with experience in industry consultancy, funded research leadership and doctoral supervision.

Research expertise

AI evaluation and benchmarking; reinforcement learning; mathematical reasoning; computational complexity; algorithm design; formal verification; information theory.

Selected AI research

Humanity's Last Exam

Contributed expert questions to the AI capability benchmark and received a US\$5,000 prize in 2025. Member of the HLE Contributors Consortium behind the benchmark paper published in Nature in 2026.

Reinforcement learning and generalisation

With Bei Zhou, investigated limitations of AlphaZero-style learning in impartial games, distinguishing strong competitive performance from mastery across game states. Published in Machine Learning (2026). Subsequent work explores multi-frame representations for learning mathematical strategy.

Mathematical models of LLM text ecosystems

Developed an exactly solvable n-gram framework separating neutral drift from normative selection in recursive text environments. Preprint, 2026.

Professional experience

Queen Mary University of London | 2000-present

Current role: Reader in Computer Science

- Conduct original research connecting mathematical structure, computational limits and learning; supervise doctoral research in complexity and game AI.
- Led funded research, including EPSRC grant EP/H016015/1 (£414,367), and co-founded and directed the Centre for Discrete Mathematics.

Fetch.AI, Cambridge | Jun 2018-Mar 2019

Research Scientist Consultant, part-time

- Developed and analysed models of consensus protocols, hard forks and blockchain governance; contributed to work on a self-organising smart ledger involving distributed AI.

Education

University of Oxford | DPhil in Mathematics, Mathematical Logic, 1994

University of Copenhagen | MSc in Mathematics and Physics, 1989

Selected research projects

Interactive theory paper and Lean proof explorer | 2026

Original mathematical research: One Unit Separates Polynomial Time from Undecidability in Term Coding. Studies a sharp boundary between polynomial-time computation and undecidability in term-coding problems with disequality constraints.

Presents the theory through an interactive paper linked to an accompanying Lean-checked formalisation. Readers can inspect formal statements, proof states and dependencies; the verification scope and external assumptions are documented.

[Interactive paper](#) | [Lean source](#)

Gomulda AI search and learning lab | 2026

Research platform combining exact graph-colouring optimisation, Monte Carlo tree search and minimax endgame search with a separate experimental neural agent trained through self-play. The browser-based Learning Lab supports training, checkpoints and agent comparisons. Based on Cristoffer Crone's strategy game.

[Live game and lab](#) | [Source code](#)

Solar Eclipse Explorer | 2026-present

Interactive scientific-computing project for the 2 August 2027 eclipse. Calculates local timings, totality duration, obscuration and the Sun's position using NASA/GSFC Besselian elements, with an animated map and sky view.

[Live explorer](#) | [Source code](#)

Selected publications and preprints

HLE Contributors Consortium, including Søren Riis. A benchmark of expert-level academic questions to assess AI capabilities. *Nature* 649, 1139-1146 (2026). [Read](#)

Bei Zhou and Søren Riis. Impartial Games: A Challenge for Reinforcement Learning. *Machine Learning* 115, article 54 (2026). [Read](#)

Søren Riis. Drift and selection in LLM text ecosystems. Preprint (2026). [Read](#)

Søren Riis. Mastering NIM and Impartial Games with Weak Neural Networks: An AlphaZero-inspired Multi-Frame Approach. Preprint, first submitted 2024; updated 2026. [Read](#)

Research leadership and communication

Doctoral supervision includes Stefan Dantchev, Sun Yun and Bei Zhou. Teaching across undergraduate and postgraduate levels, with 3,000+ students taught. Research appointments and visits include BRICS, the Fields Institute, the Isaac Newton Institute and Institut Henri Poincaré.

[Full research portfolio and publications](#)